

Univariate Data Analysis: Loan Default Prediction

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1 Introduction

The dataset used is the **Loan Default Dataset** from Kaggle, containing 148,670 samples with 28 features. The goal is **binary classification** to predict whether a loan will default (Status=1) or not (Status=0). This is a real-world financial risk assessment problem where accurate prediction can help lenders mitigate losses.

Key challenges identified initially:

- Missing values in several columns
- Suspected data entry errors in specific columns
- Highly skewed numeric features
- Mixed data types (numeric and categorical)

2 Univariate Data Analysis and Cleaning

2.1 Initial Preprocessing

- Removed six suspicious columns due to reported egregious data entry errors that spoiled the first Assignment iteration.
- Dropped all rows with missing values (9% of data removed) - lack of domain knowledge required to create a sophisticated imputer that is guaranteed not to worsen results and relatively small data loss for a simple linear model are the reason I am inclined to try chances with simple removal.
- Verified that the target distribution remained stable (default rate: 24.6% $\rightarrow$ 25.3%).
- Removed single-value columns (only **year** was dropped).

2.2 Descriptive Statistics

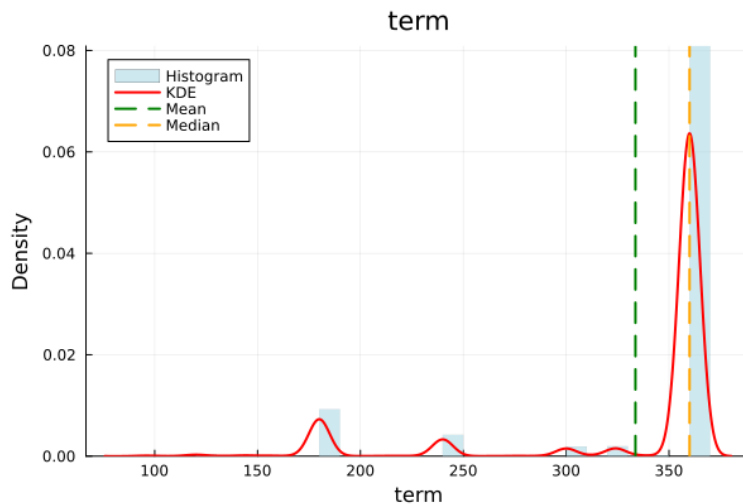
Four main numeric predictors were analyzed: **loan.amount**, **term**, **income**, and **Credit.Score**. 20+ supplementary categorical predictors are served with One-Hot Encoding.

Some findings after investigating descriptive statistics: Obviously, sociologists could make so much more information out of reading medians and means of income and such. However, the points below are useful to know for the task of improving a classifier:

- **Credit.Score**: Nearly symmetrical (skewness ≈ 0), light-tailed (bear with me until Visualization)
- **income**: Extremely right-skewed (skewness = 17.6), heavy-tailed (kurtosis ≈ 912)
- **loan.amount**: Right-skewed (skewness = 1.7), heavy-tailed
- **term**: Left-skewed (skewness = -2.1), moderate kurtosis (bear with me until Visualization)

2.3 Visualization

Histograms with KDE overlay were created for each numeric variable, showing distribution shapes, mean/median positions, and IQR boundaries.



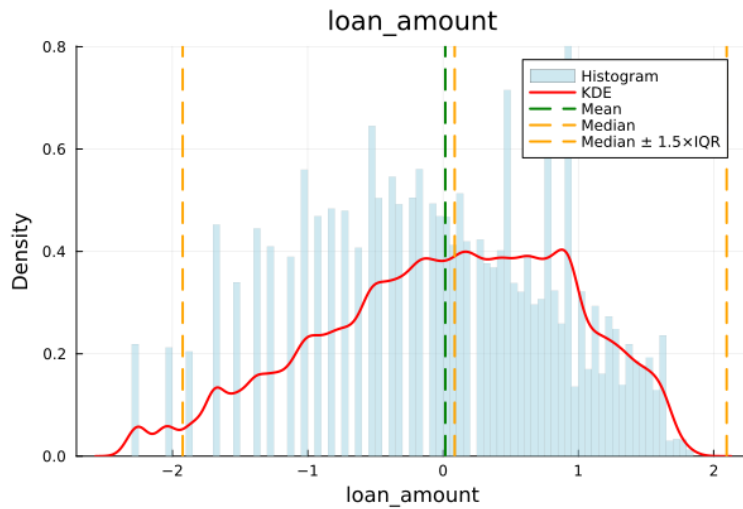
Observations:

- **Credit_Score** resembles a uniform distribution. Shall we formally test it? Probably not, given that failing to reject the uniformity hypothesis does not give us any instruments to improve a classifier.
- **income** and **loan_amount** follow LogNormal-like distributions with extreme outliers - more discussion on that topic in the notebook (though, a single phrase is worth mentioning here: <https://www.stat.cmu.edu/csha03.pdf>)
- **term** appears as a discrete variable with most values at maximum (360)

2.4 Transformations and Outlier Handling

Applied transformations:

1. **Credit_Score**: Standardized (mean=0, std=1, but it stayed uniform-like)
2. **Income**:
 - Zero-income cases replaced with median income - I did not want them to spoil the game after applying logarithm
 - Logarithm applied to the remaining and "medianized" values
 - Binary indicator **zero_income** created - I still wanted to retain the record if a person applying for a loan has no registered income
 - Standardized after log transformation
 - Trimmed 1st and 99th percentiles - more discussion on that is again in the notebook, but main idea is that extremely rich people may behave and take out loans with completely different intentions
3. **Loan Amount**:
 - Logarithm applied (In contrast to income, $\ln(0)$ was not a problem here - values of zero would be automatically expunged from the dataset indicating a data entry error)
 - Standardized
 - Trimmed 1st and 99th percentiles - motivation mostly coincides with trimming income
4. **Term**: Normalized to $[0, 1]$ range (min-max scaling with 1 getting assigned to a full-term of 360 days)



More visuals like this example are in the notebook.

Some justification:

- Financial variables typically follow log-normal distributions (have you looked at the link yet?)
- Trimming extremes removes potential data errors and increases robustness - I could take down everything that falls out of the interquartile range, but the distributions appeared benign after trimming only a single percentile from the both sides
- Normalization ensures equal scaling for gradient-based optimizers (and ability to converge for gradient-free ones!)
- Zero-income handling preserves information while avoiding $\ln(0)$ issues (already mentioned though)

2.5 Categorical Variables

- All categorical variables were one-hot encoded
- Missing values were removed
- Sometimes there is a chance to transform categorical into at least ordinals, but here it was not the case - either from an ethical standpoint (how do you even choose what gender is to be assigned what numerical value?) or purely practical (some variables tied to loan contract have no "lower" or "upper" values to each other)

3 Results

3.1 Baseline Model Performance

Model	F1 Score	ROC-AUC
Logistic Regression (no regularization)	0.365	0.552
Logistic Regression (L2 $\lambda = 0.01$)	0.599	0.822

3.2 Improved Model Performance

Model	F1 Score	ROC-AUC
Logistic Regression (after univariate analysis)	0.641	0.838

3.3 Discussion of benefits

- **Performance Improvement:** both scores visibly increased
- **Solver Compatibility:** After data grooming, Newton and LBFGS solvers became stable - convergence is now achieved much faster. Bonus: ProxGrad solver became available, which unlocked

optimization targets unrestricted by gradient and differentiability requirements - although not the point of the assignment, imagine how cool it is to be able to optimize for precision or recall directly.

- **Robustness:** Trimming extremes made the model less sensitive to outliers
- **Interpretability:** Transformed features have more meaningful scales, but that will come into play when regularization is applied or feature interactions and selections start happening.